

Instalación del servidor Tomcat

Despliegue de Aplicaciones Web

En la presente práctica tenemos como objetivo realizar y documentar lo necesario para la puesta en marcha de un servidor de aplicaciones usando Tomcat sobre Ubuntu Server. Necesitamos para ello realizar la instalación y configuración del software necesario:

1. Instala el servidor Tomcat 11
2. Configura y fija para el sistema las variables de entorno para la correcta ejecución de Tomcat.
3. Gestiona el servicio instalado:
 - a. Comprueba el estado, para, inicia.
 - b. Deshabilita el arranque automático, reinicia la máquina y comprueba el estado.
 - c. Habilita, inicia y comprueba el estado.
4. Investiga sobre la estructura de gestor directorios de Tomcat en las dos opciones de instalación indicadas y explica brevemente que contiene cada directorio de Tomcat.
5. En el terminal, averigua los puertos TCP y/o UDP que están a la escucha. Cambia el puerto en el que escucha Tomcat, reinicia el servicio y vuelve a comprobar los puertos.
6. Mediante el navegador accede al servidor. Comprobar los logs del mismo.
7. Instala la aplicación **sample.war** que se adjunta en el enlace.
8. Instala de forma manual en `/var/lib/tomcat11/webapps` la aplicación **hello-world.war** que se adjunta en el enlace.

```
root@ip-172-31-39-65: ~
root@ip-172-31-39-65: ~ 80x24
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-39-65:~# readlink -f $(which java)
/usr/lib/jvm/java-25-openjdk-amd64/bin/java
root@ip-172-31-39-65:~# ^C
root@ip-172-31-39-65:~# apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Reading package lists... Done
root@ip-172-31-39-65:~# nano /etc/environment
root@ip-172-31-39-65:~# source /etc/environment
root@ip-172-31-39-65:~# echo $JAVA_HOME
/usr/lib/jvm/java-21-openjdk-amd64
root@ip-172-31-39-65:~#
root@ip-172-31-39-65:~# apt-get install tomcat9 tomcat9-admin tomcat9-docs tomcat9-examples -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package tomcat9
E: Unable to locate package tomcat9-admin
E: Unable to locate package tomcat9-docs
E: Unable to locate package tomcat9-examples
root@ip-172-31-39-65:~#
```

```
root@ip-172-31-39-65: ~
root@ip-172-31-39-65: ~ 80x24
● tomcat10.service - Apache Tomcat 10 Web Application Server
   Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset:~>
   Active: active (running) since Thu 2026-04-30 10:19:26 UTC; 55s ago
 Invocation: e2efbc9c712649d6a65b86d1647cd828
    Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
   Process: 4718 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (c~>
   Main PID: 4723 (java)
      Tasks: 29 (limit: 627)
     Memory: 101.9M (peak: 103.9M)
        CPU: 8.732s
    CGroup: /system.slice/tomcat10.service
           └─4723 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.conf~>

Apr 30 10:19:31 ip-172-31-39-65 tomcat10[4723]: Deployment of deployment descri~>
Apr 30 10:19:31 ip-172-31-39-65 tomcat10[4723]: Deploying deployment descriptor~>
Apr 30 10:19:31 ip-172-31-39-65 tomcat10[4723]: The path attribute with value [~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: At least one JAR was scanned fo~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: Deployment of deployment descri~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: Deploying web application direc~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: At least one JAR was scanned fo~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: Deployment of web application d~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: Starting ProtocolHandler ["http~>
Apr 30 10:19:32 ip-172-31-39-65 tomcat10[4723]: Server startup in [4105] millis~>
lines 1-23
```

It works !

If you're seeing this page via a web browser, it means you've setup Tomcat successfully. Congratulations!

This is the default Tomcat home page. It can be found on the local filesystem at: `/var/lib/tomcat10/webapps/ROOT/index.html`

Tomcat veterans might be pleased to learn that this system instance of Tomcat is installed with `CATALINA_HOME` in `/usr/share/tomcat10` and `CATALINA_BASE` in `/var/lib/tomcat10`, following the rules from `/usr/share/doc/tomcat10-common/RUNNING.txt.gz`.

You might consider installing the following packages, if you haven't already done so:

tomcat10-docs: This package installs a web application that allows to browse the Tomcat 10 documentation locally. Once installed, you can access it by clicking [here](#).

tomcat10-examples: This package installs a web application that allows to access the Tomcat 10 Servlet and JSP examples. Once installed, you can access it by clicking [here](#).

tomcat10-admin: This package installs two web applications that can help managing this Tomcat instance. Once installed, you can access the [manager webapp](#) and the [host-manager webapp](#).

NOTE: For security reasons, using the manager webapp is restricted to users with role "manager-gui". The host-manager webapp is restricted to users with role "admin-gui". Users are defined in `/etc/tomcat10/tomcat-users.xml`.

401 Unauthorized

You are not authorized to view this page. If you have not changed any configuration files, please examine the file `conf/tomcat-users.xml` in your installation. That file must contain the credentials to let you use this webapp.

For example, to add the `manager-gui` role to a user named `tomcat` with a password of `s3cret`, add the following to the config file listed above.

```
<role rolename="manager-gui"/>
<user username="tomcat" password="s3cret" roles="manager-gui"/>
```

Note that for Tomcat 7 onwards, the roles required to use the manager application were changed from the single `manager` role to the following four roles. You will need to assign the role(s) required for the functionality you wish to access.

- `manager-gui` - allows access to the HTML GUI and the status pages
- `manager-script` - allows access to the text interface and the status pages
- `manager-jmx` - allows access to the JMX proxy and the status pages
- `manager-status` - allows access to the status pages only

The HTML interface is protected against CSRF but the text and JMX interfaces are not. To maintain the CSRF protection:

- Users with the `manager-gui` role should not be granted either the `manager-script` or `manager-jmx` roles.
- If the text or jmx interfaces are accessed through a browser (e.g. for testing since these interfaces are intended for tools not humans) then the browser must be closed afterwards to terminate the session.

For more information - please see the [Manager App How-To](#).



Gestor de Aplicaciones Web de Tomcat

Mensaje: OK

Gestor

[Listar Aplicaciones](#) [Ayuda HTML de Gestor](#) [Ayuda de Gestor](#) [Estado de Servidor](#)

Aplicaciones

Ruta	Versión	Nombre a Mostrar	Ejecutándose	Sesiones	Comandos
/	Ninguno especificado		true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/docs	Ninguno especificado	Tomcat Documentation	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/examples	Ninguno especificado	Servlet and JSP Examples	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/host-manager	Ninguno especificado	Tomcat Host Manager Application	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos
/manager	Ninguno especificado	Tomcat Manager Application	true	1	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar ≥ 30 minutos

Desplegar

Desplegar directorio o archivo WAR localizado en servidor

Trayectoria de Contexto (opcional):

Version (for parallel deployment):

URL de archivo de Configuración XML:

URL de WAR o Directorio:

Archivo WAR a desplegar